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PI Name:	Mason, Christopher Ph.D.		
Project Title:	Leveraging the Microbes of Earth's Extreme Environments for Sustainable Plant Growth in Lunar Regolith		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
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Space Biology Cross-Element Discipline:	None		
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PI Email:	chm2042@med.cornell.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	203-668-1448
Organization Name:	Weill Medical College of Cornell University		
PI Address 1:	1305 York Ave, Room Y13-04, Box 140		
PI Address 2:			
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10021-5663	**Congressional District:**	12
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Contact Monitor:	Blancaflor, Elison B.	**Contact Phone:**	321-501-5368
Contact Email:	elison.b.blancaflor@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	No change; the same teams and researchers are present.		
COI Name (Institution):	Church, George Ph.D. (Harvard Medical School) Ferl, Robert Ph.D. (University of Florida, Gainesville) Henriksen, James Ph.D. (Colorado State University) Paul, Anna-Lisa Ph.D. (University of Florida, Gainesville) Peebles, Christie Ph.D. (Colorado State University) Tierney, Braden Ph.D. (Weill Medical College of Cornell University)		
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Task Description:	Modulating the plant-microbe axis will be critical for sustainably cultivating crops in space, which itself is a critical step on the path to sustaining human life off of Earth. However, the stressors of space and its lack of inherent resources for plant growth make this a challenging goal. Therefore, it is necessary to identify how materials immediately on hand during spaceflight, like human waste, lunar regolith, or microorganisms, can be used to support plant life with minimal use of Earth-derived resources. Successfully doing so will require modulating all dimensions of plant health, including plant microbiota that play a critical role in stress tolerance. Here we propose using the microbiology of Earth's extreme, lunar-relevant environments to identify microbial biostimulant consortia that enable plant growth in synthetic lunar regolith. This effort takes advantage of our team's ongoing work building a "living database" from geochemically diverse areas on Earth. By the time the work laid out here will begin, this resource will >1000 sequenced samples, including preserved environmental material (e.g., sediment), microbial consortia enriched in different media types (e.g, high carbon dioxide) , and single isolates, including many novel Cyanobacteria. Notably, many of these Cyanobacteria produce exopolysaccharides (EPS), which, in the presence of plant roots, can increase soil water retention, plant desiccation tolerance, and other plant stress resistance. Through this existing dataset, as well as an additional expedition to the Cima Volcanic Field -- a geochemically "lunar-relevant" desert site in Southern California -- we will identify defined microbial consortia that enhance plant growth in lunar regolith. Our team comprises a diverse set of leaders in the fields of crop cultivation in space, industrial agriculture, computational biology, DNA/RNA sequencing, and extreme microbiology. We will integrate these skillsets to test, via high-throughput screening, how Arabidopsis thaliana health (as measured by total plant weight, growth rate, and leaf transcriptomics) varies as a function exposure to different environmental material, geochemically enriched consortia, and EPS-producing Cyanobacteria. We will additionally test if human waste analogs can be used to enhance the biostimulant effects of different microorganisms. Through DNA-sequencing and a bioinformatic association study with growth outcomes, we will identify candidate minimum sets of microbes that confer tolerance to synthetic lunar regolith in the presence or absence of human waste. In total, this work will advance NASA's goal of sustaining life in space and on other planets via efficient crop cultivation with minimal reliance on Earth-based resources.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This effort will further NASA's goal of long-term survival in space, and in exploration habitats, by advancing in situ use of regolith for plant growth using targeted plant microbiomes. This also has a broader application in stimulating plant growth in stressful terrestrial environments.
	Original Aims, Grant Objectives, and Research Summary Increasing crop growth in the space environment is critical for human survival off of Earth. This requires modulating all aspects of plant physiology, including their native microbiota, which play a critical role in stress tolerance. In this Plant Research Investigation, we propose identifying beneficial microbial consortia from Earth’s extreme environments with lunar-like geochemistries that are capable of protecting and promoting plant health in simulated lunar regolith. This effort will take advantage of a substantial existing dataset and also involve collecting further samples. Our team – a joint effort between the Two Frontiers Project (twofrontiers.org) and a number of additional collaborators – has recently executed multiple expeditions to extreme environments in order to identify microbes capable of supporting human life in space. We use the geochemistry of the natural environment to identify growth conditions for target organisms of interest – collecting, when possible, sediment/soil, water, and biomass (e.g., biofilm) samples and exposing them to different enrichment conditions to select for microbial phenotypes of interest. With successfully completed missions to volcanic seeps in the Aeolian islands and a variety of sites in the Colorado Rocky Mountains (including forested, fire scarred, urban, and plant associated environs), and an additional trip to high CO2 springs in Colorado completed before this proposed work would begin, we will have built a “living database” of >1,000 sequenced and cryopreserved environmental samples, microbial consortia enriched in high-CO2 media, and single isolates. We have enriched for desiccation resistant cyanobacteria that produce exopolysaccharides (EPS), a mechanism of increasing soil water retention, plant desiccation tolerance, and other plant stress resistance. Preliminary results from 11 unialgal cultures show desiccation tolerance and many exhibit high EPS production. With work proposed here, we will both leverage and extend this existing resource to identify microbial consortia that support plant growth in lunar-like environments, thereby minimizing the amount of Earth-derived material required to cultivate crops. We will, in parallel, conduct an expedition to the Mojave desert to collect and enrich for plant-associated and desert-crust microbes capable of tolerating lunar regolith and the space environment. We will then screen these enriched consortia, as well as our existing banked samples, against Arabidopsis thaliana grown in varying soil using metagenomic data to prioritize and interpret experiments. Overall, by devising microbial solutions to minimize our reliance on Earth-based materials when cultivating crops, we will identify candidate defined consortia for stimulating plant growth in stressful environments – including lunar regolith. This effort will further NASA's goal of long-term survival in space and in exploration habitats, including those envisioned for the Highlands of the lunar south pole. Specific Aims Aim 1: Collect samples from the Cima Volcanic Field and enrich in media designed to optimize growth in the space environment Using our existing methodologies for collecting, sequencing, and culturing samples from extreme environments, we will collect soil and rhizosphere associated microbes of a variety of desert plants and soil crusts. We will sample the area surrounding Cima Volcanic Field in the Eastern Mojave desert, which is: (1) nutrient poor, (2) alkaline, and (3) silicon-rich. These characteristics make this site uniquely relevant to plant growth in lunar regolith and therefore the aims of this proposal. Furthermore, the volcanic rock formations there are composed of minerals including anorthosite, an analogue to lunar regolith. We will use the well-characterized geochemistry of this site to design custom media for lunar-relevant microbial enrichments. Our team has decades of cumulative experience in carrying out successful microbial/genetic expeditions, including logistics such as permitting, field data and sample collection, and transport. We will leverage our existing protocols to collect and bank a total of 100 soil and rhizosphere samples from diverse plant life and topsoil locations. Each sample will be split, with half being stored on dry ice in glycerol and dimethyl sulfoxide (DMSO) and half immediately being placed into an enrichment media design based on geochemistry of Cima. Enrichments and samples will be banked at Weill Cornell Medical College. Aim 2: Identify a defined microbial consortia for regolith-associated biostimulant properties We will inoculate A. thaliana seeds with different environmental and enriched biomass (enriched for microbial phenotypes and geochemical characteristics of interest) from the three expeditions. We will have banked a total of 800 enriched consortia; 50 of these will be selected based on growth, microbial composition, and functional genetic repertoire (from shotgun sequencing) and

Task Progress:	the geochemical signatures of their source environment. Over a four-week window, these seeds will be grown and inoculated in triplicate in Lunar Highlands regolith simulant to approximate the in situ resources of the Artemis base. We will additionally spike-in minimal human waste analogs for some replicates to determine if synergistic effects between microbes, plants, and waste products can be identified. Regolith will be non-sterile in order to simulate realistic conditions. Full plant biomass will be measured at the end of each experimental window, and overall growth rate will be ascertained. In accordance with our University of Florida (UF) team's past efforts, RNA-sequencing on leaf tissue will be used to ascertain an additional readout of stress. The external and endophytic microbiota from high and low growth rate/extent plants will be characterized with shotgun sequencing. Microbes associated with growth will be bioinformatically identified and their enrichment and isolation from stocked screening material will be attempted. Individual isolates will be tested alone and in different combinations to determine a minimal set of organisms required for maximal benefit. Overall progress and work to date We have accomplished the FY25 Aims from our initial proposal and are on track (or ahead) in all dimensions to complete our work. Specifically, we have fully completed Aim 1, in line with our proposed work timeline, and begun work on the goals of Aim 2. We additionally have completed pilot experiments relevant to the coming planned efforts. Specific accomplishments include the following: 1. We used our research permit (MOJA-2024-SCI-0042) to take the battery of samples needed for completion of the study from the Cima Volcanic Field within the Mojave Natural Preserve in 2024, and processed, banked, and tested these samples in various culture mediums. 2. Samples were aliquoted, banked, and QC'd for long-term storage and metagenomics processing. 3. The 76 final set of samples collected in the Cima Volcanic Field, as well as controls, were extracted, sequenced, and analyzed for microbial characterization, including taxonomic profiles from shotgun sequencing, de novo assembly for metagenome-assembled genomes (MAGs). 4. We updated our bioinformatic pipelines for identifying unusual taxonomies and exopolysachharide-associated (and other potentially beneficial) functions in metagenomic data. This work led to two papers, which highlight multi-domain mapping tools for species identification and likely functional impact: a. "CAMP: A modular metagenomics analysis system for integrated multi-step data exploration." https:// . This paper was accepted for publication in December 2025 and will be published in 2026. b. "The Microbe Directory: a centralized database for biological interpretation of microbiome data." We also compared our microbial annotations with annotations collected by artificial intelligence (AI) and demonstrate that, despite the high speed of AI in reviewing and collecting microbial data, annotation requires domain knowledge and therefore manual curation. https:// 5. We continued culturing experiments on Cyanobacteria that may be relevant to crop protection in lunar regolith. Specifically, we further characterized several species which produce exopolysachharides from additional soil crust samples outside of the Mojave Reserve. 6. We have begin comparing metagenomics data from a parallel project, called the Microbe Mineral Atlas, which will be helpful as comparisons to other extremophiles: https:// . This includes other sites from deserts, mines, and high radiation environments. Awards, publications, presentations (Ed. Note: See Cumulative Bibliography) 1. Sierra MA, Ryon K, Arikatla MR, Elshafey R, Bhaskar H, Proszynski J, Bhattacharya C, Shaaban H, Danko DC, Ambrose P, Spaulding SA, Zambrano MM, Consortium TMD, Mason CE. The Microbe Directory: a centralized database for biological interpretation of microbiome data. Database (Oxford). 2025 Jan 18;2025:baaf060. doi: 10.1093/database/baaf060 . PMID: 40996703; PMCID: PMC12462379. 2. Mak L, et al., CAMP: A modular metagenomics analysis system for integrated multi-step data exploration. In press. NAR Genomics and Bioinformatics. New Technology 1. Updated taxonomic data was entered into the CAMP paper. This will be publicly released for use by the community. Future Research Plan In the coming year, we plan to: 1) Sequence the last of the samples from the collections. 2) Continue comparative analysis across the sites from our project. 3) Apply for real regolith samples for testing our candidate species. Significant Budget or Key Personnel Changes N/A Study Progress Study is on course and no changes are proposed.
Bibliography Type:	Description: (Last Updated: 07/13/2026)

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